FTIR versus Raman spectroscopies: theory and applications

Abstract

From the classical principles of electromagnetic scattering to the omnipresent modern techniques such as ATR-FTIR, this comparative study will delve into the broad theoretical and practical differences and similarities between Fourier-Transform Infrared (FTIR) spectroscopy and Raman spectroscopy.

We will explain the capabilities of each tool, what kinds of materials can be studied with them, in what fields they are the most or least useful, the way they complement each other, and perhaps most importantly what the theoretical underpinnings of those capabilities are.

Introduction

The first step in chemistry and material science is to know what chemicals or materials make up the samples that are being used or studied. Physicists have developed a variety of techniques to characterise most known substances whether they are arranged in solids, liquids, gases, etc.

One of the most common and useful ways to do this is using **spectroscopy** which typically consist of sending some light on the sample and analysing the transmitted, emitted, or scattered light to yield a spectrum of intensity as a function of wavenumber (in cm^{-1}).

Once again, many different spectroscopy techniques exist, but two stand out. These are regrouped under the umbrella term of « **vibrational spectroscopy** »[1] because they probe the intrinsic vibrational or rotational modes in the sample.

Raman spectroscopy, stemming from the seminal work of Sir Chandrasekhara Venkata Raman in 1928, identifies materials through inelastic scattering of light. It discerns the vibrational and rotational[5] fingerprints of molecules by scrutinizing the oscillations in their polarizability[2]. Conversely, **FTIR spectroscopy**, tracing its roots to the mid-20th century, operates on the principle of molecular vibrations' absorption of infrared radiation and the mathematical power of the Fourier-Transform. With its advanced variants such as Attenuated Total Reflectance (ATR), FTIR has revolutionized the field of spectroscopy by enabling analysis without the need for laborious sample preparation, making it a standard technique in numerous scientific domains.

Although both techniques serve as indispensable tools to identify molecules with a high degree of confidence and precision, **their applications diverge significantly** ultimately because of those differing physical principles. Raman spectroscopy is at an advantage for studying inorganic systems and crystals while FTIR is good at determining functional groups in organic molecules. It is ultimately to understand these differences and their origin that we write this comparative analysis.

Methods

This study is separated into the **theoretical** and **practical** parts. In each part we try to be methodical in explaining the basics of the techniques and then comparing them along several variables.

Thus, we will first concentrate on understanding the theory behind the Raman effect step-by-step followed by the simpler process of IR absorption. With this foundational basis, we can compare the theoretical advantages and limitations of each technique on a set of parameters such as:

selection rules, feasibility of observations, sensitivity, acquisition time, and spatial resolution.
Then, we explain how that translates into the practical use cases for each tool and how some of those theoretical limitations can be overcome in practice: which kinds of materials are best to study with Raman or FTIR and which fields are the most affected by those differences.

Raman vs FTIR: theory

Before any comparison can be made, let us begin our investigation by understanding the physical principles behind each technique.

The Raman effect

The Raman effect is at its heart a type of inelastic scattering of light based on oscillations of polarizability of a molecule due to its vibrations or rotations. In this section we will explain step-by-step what this means.

Elastic scattering

Let us begin with **elastic scattering**. When light is shone on an object or a molecule, it can « bounce back » in a different direction. This is what we call scattering. Before the 20th century, scientists thought that scattering of light was elastic: that it should not lose nor gain energy in the process of « bouncing back ». This type of scattering is called **Rayleigh scattering**[2].

In the framework of classical electromagnetism, we can understand this effect like this: when an electromagnetic wave hits a charged particle like an electron, the electric field component of the wave exerts an alternating force on the particle creating an alternating acceleration and thus a vibrating movement. This vibration has the exact same frequency as the incident wave. Because the accelerating charge produces electromagnetic waves, a secondary wave with the same frequency is emitted by the electron that we call the **scattered wave**.

In the framework of quantum mechanics, the electromagnetic wave is a collection of photons of energy $\boldsymbol{E=h\omega}$ where ω is the frequency and h the Planck constant. This means that for Rayleigh scattering, the scattered wave also has the same energy as the incident wave: the process is labelled elastic in the same way that a bouncing elastic rubber ball retains all its energy after the bounce.

Inelastic scattering: the history of the Raman effect

Adolf Smekal predicted the **inelastic scattering** of light in 1923[13] while from 1922 to 1928, Indian physicist Sir Chandrasekhara Venkata Raman and his collaborators published a series of works shortly after his PhD from the University of London. This culminated in the discovery of the Raman effect on February 16th of 1928 in organic liquids: he showed that when monochromatic light was shone through some molecules, some photons would scatter with a slightly different energy than the incident photons, revealing frequencies of molecular vibrations in the process. This was a revolutionary discovery that earned him the 1930 Nobel Prize in physics.[14]

Raman scattering turns out to be a very weak effect. Out of 10 million incoming photons, one can usually expect 1 000 to 10 000 Raileigh-scattered photons but only 1 Raman-scattered photons.[2]
It also has a very similar frequency to the incident light, making it indispensable to use light of a single well-defined frequency as the source **(monochromatic light)**. Raman and other physicists had to use mercury lamps for that and used photographic plates as detectors.[15] The introduction of lasers in the early 1960's and CCD detectors in the late 1960's changed everything. The technique was then greatly refined until it was recognised in 1998 as « International Historic Chemical Landmark » by the American Chemical Society.[24] This technique has now blossomed into a multitude of widely used variations such as surface-enhanced Raman, resonance Raman, tip-enhanced Raman, and many more.

The exact mechanism of Raman scattering involves the oscillations of polarizability within the molecules or lattices studied. In the next sections we explain the concept of polarizability and its relationship with the vibrations of atomic bonds.

Polarizability

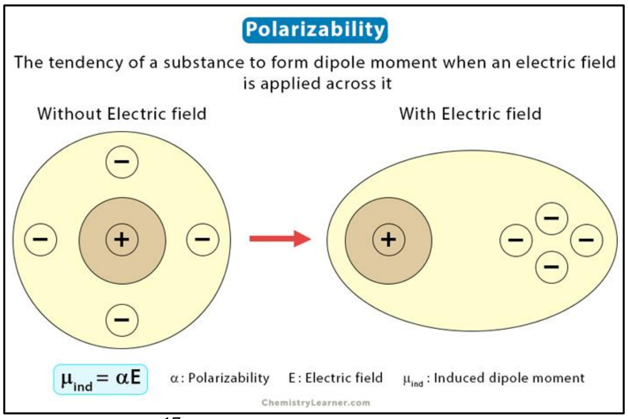

Figure 1[17]: exaggerated schematic view of polarizability

To explain this concept, let us take the example of a hydrogen atom. When an electric field is applied to it, the atom becomes polarized: schematically we can say that the electron goes one way and the nucleus the other way due to the Coulomb force acting on the charges (although the movement of the nucleus is small compared to that of the electron due to its much higher mass). This separation induces a dipole moment[1] or a polarization $\boldsymbol{\mu} = q\boldsymbol{d}$ where q is the magnitude of the charges being separated and d is the distance between the charge centers.[17]

This polarisation effect is present in every substance or molecule. The degree to which a substance can be polarized by an applied electric field E is called the polarizability α. One can make a good first order approximation[13]:

$$\mu = \alpha . E$$

In the most general context, E is plugged in the formula as a vector, α becomes the polarizability matrix and μ is the 3d polarization vector.[13]

Many parameters can affect polarizability. The more a nucleus can hold electrons in a tight orbital around it, the harder it is for a field E to induce a polarization between them. This means that less electronegative atoms with a larger radius tend to have a higher polarizability. More complex non isotropic effects take place for certain molecules[17] which also justifies the use of a polarizability matrix with non-zero diagonal coefficients.

More importantly, it can be easier to polarize a bond that is mechanically in a stretched position. A famous example for this would be the symmetric stretch of a CO_2 molecule.[20]

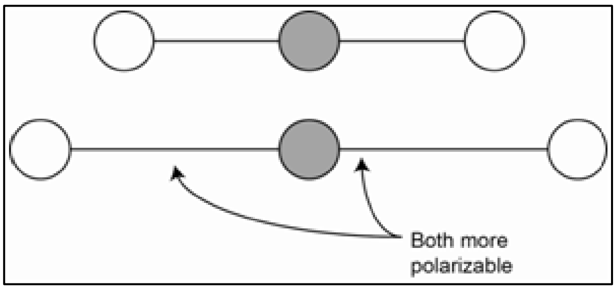

Figure 2: Representation of length-dependent polarizability

Vibration-modulated polarizability

The first important point to understand the Raman effect is that, as we saw, the polarizability dictates the **magnitude of the dipole moment** $\boldsymbol{\mu}$ induced by the light wave, and as a result also dictates the **strength of the reemitted (scattered) wave**.

The second important point is that molecular motions such as stretching of bonds can modify polarizability and thus some vibrational motions can cause a slight **oscillation of polarizability** (with the same frequency). A first order approximation can again be made for the relationship between the polarizability tensor and the vibrational coordinates Q.[13]

$$\alpha_{ij} = (\alpha_{ij})_0 + (\frac{\partial \alpha_{ij}}{\partial Q_k})_0 Q_k$$

For example, the CO_2 molecule is more easily polarized when it is in a symmetrically stretched position. Therefore, the linear symmetric stretching mode of vibration of the CO_2 molecule will produce oscillations of α and is said to be **Raman-active**. In general, any system in which $\frac{\partial \alpha_{ij}}{\partial Q_k} \neq 0$ is said to be Raman-active. This criterion is called the « **selection rule** » of Raman spectroscopy.[16]

Putting the two points together: because vibrational motion is a lot slower than the frequencies of light used in Raman, a light wave arriving on a Raman-active bond will trigger the emission of a scattered wave with an amplitude modulated periodically by the molecule's vibrations. In terms of dipole moment, this signal will take the overall form[13:]

$$\mu = \alpha_k . E_0 \cos(\omega_0 t) = \alpha_0 . E_0 \cos(\omega_0 t) + (\frac{\partial \alpha_k}{\partial Q_k})_0 . E_0 Q_{k0} \cos(\omega_0 t) \cos(\omega_k t + \delta_k)$$

Where ω_0 is the frequency of light and ω_k is the much lower frequency of molecular vibrations. The first term corresponds to elastic Raileigh scattering and the product of cosines corresponds to the Raman wave modulated by the vibrational frequency.

When decomposed into its component frequencies, such a modulated signal will turn out to be composed of photons with frequencies $\omega_0 + \omega_k$ and $\omega_0 - \omega_k$. This can be understood using the trigonometric formula:

$$\cos(a)\cos(b) = \frac{1}{2}\cos(a+b)\cos{(a-b)}$$

Photons of lower energy are labelled as Stokes while those of higher energy are called anti-Stokes.[2] This is the origin of non-elastic scattering or Raman scattering. We can see that the frequency shift of the scattered Raman photons is directly the same as the characteristic frequency of vibration of the molecule which we are investigating.

Energy states

Up until now we have mostly described this effect with a classical electromagnetism point of view. Here is the associated quantum point of view (i.e.: the Raman effect in term of discreet energy transitions).

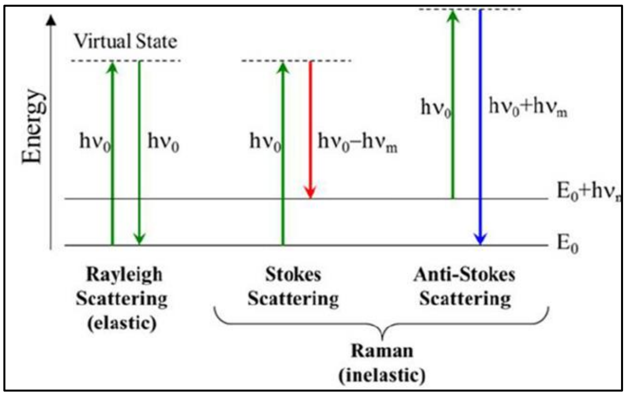

Figure 3: Jablonski energy diagram of Raman processes

In this simplified Jablonski energy diagram[18] are shown the three different kinds of processes that we have seen so far as transitions between intrinsic energy states in a molecule. Higher horizontal lines mean higher energy states. The two lower lines on the diagram are the first two energies of vibrational states in the ground electronic state. Excited electronic states possess a much higher energy than the vibrational ones and are not shown here. Arrows show absorption (up) or emission (down) of photons.

Raman spectroscopy uses laser light in the near IR to near UV region: these photons have too much energy to induce direct vibrational transitions, but too little to induce electronic transitions. Therefore, they are said to excite molecules to an intermediate « virtual » vibrational state before relaxing to a real vibrational state, emitting the scattered photons in the process.[1] The relaxed state may be directly above or below the initial one resulting in Raman shifts of the same magnitude as vibrational transition energies.

Virtual states are not actual intrinsic energy states of the system but instead are very short-lived transient states that are not observable. In the framework of quantum mechanics, it is a state that is not the eigenfunction of any operator, and thus no measurement will ever show one to be occupied.

To conclude this discussion, it must be mentioned that at room temperature, the excited vibrational states are by far the least common and therefore anti-Stokes scattering is typically not recorded in instruments because it needs an already excited state to start.[2]

Raman effect: conclusion and instrument setup

The Raman effect is the very slight shifting in the frequency of light when it scatters off a polarizability-changing molecular bond vibration. These bonds are known as Raman-active.

The equality between Raman shift and molecular vibration's frequencies allows us to easily interpret the spectrum and identify which Raman-active bonds are present in the sample and in what kind of arrangement. Vibrational information can also be traced back to important variables such as temperature[8] and pressure in samples.[7]

In terms of setup, the light source must be a narrow laser.[2] The Raman shifted photons can be separated from the Raileigh photons with a filter, made into a spectrum with a diffraction grating, and collected with a detector such as a CCD.[2]

Overall, the schematic setup of a Raman spectrometer and the spectrums we get at the detector look like the following.[2]

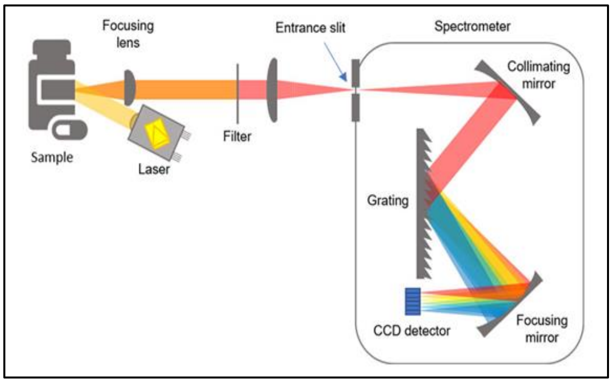

Figure 4: Setup of a Raman spectrometer[19]

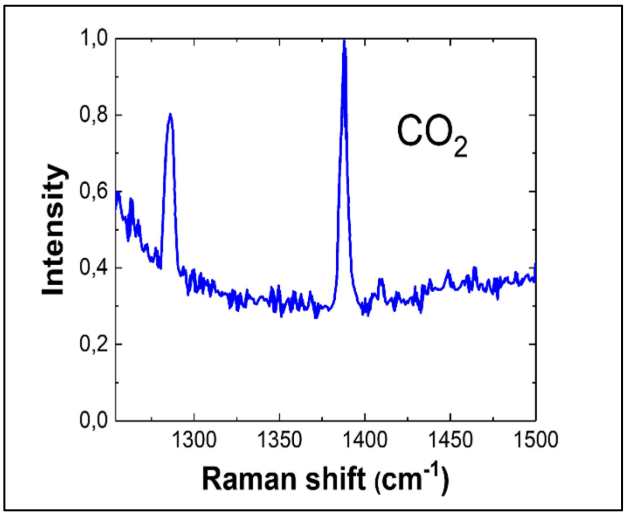

Figure 5: Raman spectrum of CO_2

FTIR spectroscopy

The full name of FTIR is Fourier-Transform Infrared absorption spectroscopy. The FT part of FTIR will be covered later, after we understand the basics of IR spectroscopy.

IR absorbance spectroscopy

The idea of IR spectroscopy is similar to that of Raman: somehow exciting molecular vibrations of a sample with light in order to get that information back in the collected light's spectrum. However, the physical principles behind the two techniques are different.

The basic principle of absorption spectroscopy is to shine a continuous spectrum of different wavelength of light through a sample and observe at the other end how the spectrum is absorbed by it. Many processes can result in the absorption of light of certain frequencies.

IR photons have the particularity of having a smaller amount of energy compared to visible or UV photons. These low energy photons can directly enter in resonance with vibrational/rotational modes in the sample and get absorbed in the small energy transition between those states. This comes in contrast to Raman photons which can't be absorbed as they are forced to relax from transient high energy virtual states.

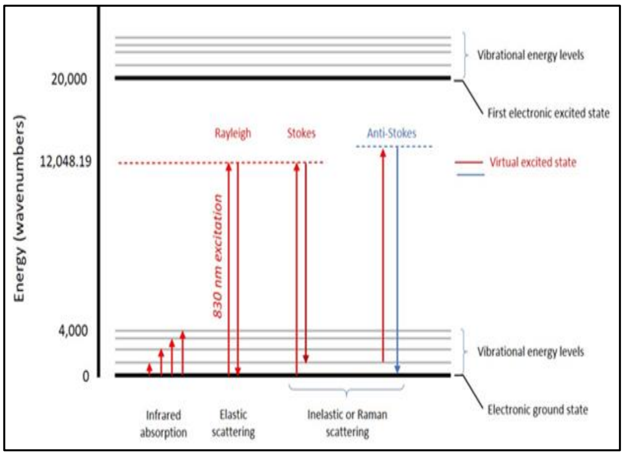

Figure 6: Addition of the IR absorption process on the Jablonski diagram

The process of IR absorption is now shown on this energy diagram alongside the Rayleigh and Raman processes[2].

IR selection rule

Just like for Raman spectroscopy, IR spectroscopy also can't excite any kind of vibrational/rotational system but only those which we will call « IR-active ». As a reminder, in the case of Raman, the selection rule was that $\frac{\partial \alpha}{\partial Q} \neq 0$. The IR selection rule is the following: a system is IR-active if its dipole moment changes during vibrations. That means that $\frac{\partial \mu}{\partial Q} \neq 0$.[16]

This comes from the fact that the IR light must enter in resonance with the molecular vibration to be absorbed. The motion of the molecule must match the way in which the IR light would make the molecule move, and the only way that such an alternating electric field can move those atoms is by displacing the partial charges in the opposite direction: to create a periodic dipole moment.
In the example of the linear molecule CO_2, the oxygens hold partial negative charges, but the molecule has no dipole moment at rest. An IR light wave arriving perpendicular to the molecule would make those oxygens move in one direction while the central partial positive carbon moves in the opposite direction, resulting in an asymmetric stretch. Thus, IR light can enter in resonance with the asymmetric but not with the symmetric stretch vibration (see Figure 7, upper right)[3].

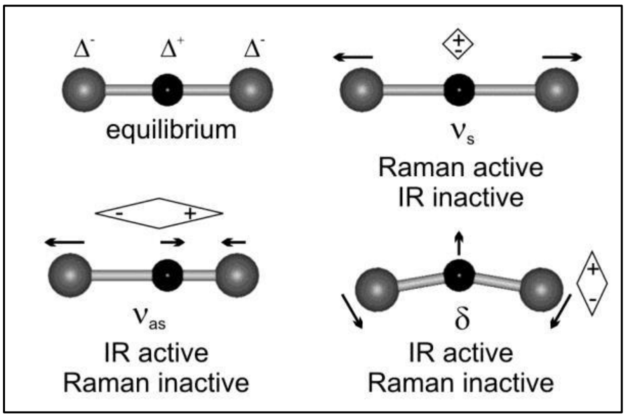

Figure 7:IR and Raman selections rules applied to CO_2

Here (Figure 7) is a comparison between the IR selection rule and Raman selection rule for the CO_2 molecule.

If one plots the percentage of light that is absorbed or transmitted as a function of wavelength (or more commonly wavenumber in cm^{-1}), he will obtain so called absorption peaks corresponding to the natural frequencies of IR-active modes.

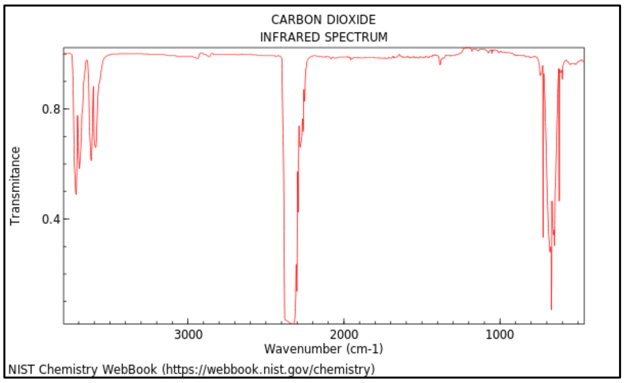

Figure 8: IR spectrum of CO_2

The above picture (Figure 8) is the IR transmittance spectrum of carbon dioxide. It shows the absorbance due to its asymmetric stretching mode but not to its IR inactive symmetric mode.

Fourier-transform IR spectroscopy

FTIR stands for Fourier-Transform Infrared Spectroscopy. IR spectroscopy became a popular technique in the 1950's while FTIR became widely adopted shortly after in the late 1960's. It then quickly became the most commonly used type of IR spectroscopy. It is now used all over the world as a standard chemistry lab technique.[25]

The difference lies in the method of spectrum generation: how we separate light into its component frequencies. In Raman spectroscopy we physically separate photons of different wavenumbers using a diffraction grating. Instead of doing this, FTIR uses an **interferometer** to obtain an interferogram of the light coming from the sample. This signal is then **Fourier-Transformed** using a computer. This allows the instrument to have better frequency resolution (sharper peaks) and have a better signal to noise ratio.

Overall, the schematic setup of a Fourier-Transform IR spectrometer looks like the following picture.[21]

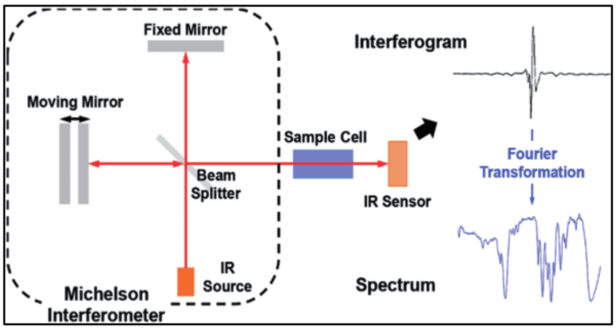

Figure 9: setup of a Fourier-Transform IR spectrometer

Comparing the two techniques

In his section we summarize and compare the theoretical advantages and inconveniences of each technique.

Theoretical inconveniences of Raman scattering

The first and most obvious disadvantage of Raman spectroscopy lies in the impossibility to detect Raman-inactive modes.
Beyond that, IR spectroscopy is based on a much simpler and much more probable physical process compared to Raman spectroscopy. Indeed, Raman scattering events have a tiny probability of occurring compared to absorption of IR light trough a sample as we can see in the transmittance spectrum of CO_2.

This usually results in much longer acquisition time needed to acquire a decent signal. So, it is difficult to monitor fast chemical reactions in real time as opposed to FTIR spectroscopy.
Stronger lasers also must be used as a result, inducing possible alterations of fragile samples (especially for microscopy) and posing a challenge in terms of the precision of the filtering setup.

On a more applied note, it turns out that IR spectroscopy is more useful in organic chemistry in dry conditions because it is great at identifying functional groups.

Finally, the use of a physical monochromator instead of an interferometer with FT means a decreased spectral resolution and more noise compared to FTIR.

Theoretical inconveniences of FTIR

Again, the most obvious inconvenient of FTIR is that in can only detect IR-active systems. Those can include very basic molecules such as all noble gases or homoatomic diatomic molecules (O_2, N_2, H_2...)[6].

Although FTIR is very sensitive especially in the detection of organic molecule's functional groups[1], it can sometimes be too much as in the case of water. The H_2O absorbance can hide the signal from other molecules beneath its own broad absorption peaks.[1] This makes it difficult to do chemistry without drying the sample or using a different solvent and is especially an issue in the case of organic chemistry and biology.

Another disadvantage of FTIR is that light must go through the sample before being observed, which is possible for gases or liquids but makes it very hard to prepare and study opaque solids. This also means that in many cases it is not possible to analyse substances remotely from one single side as it can be the case in Raman analysis (as we will see). We will see that ATR-FTIR can overcome this obstacle and make sample preparation almost trivial as it is the case in Raman analysis.

The last inconvenient of FTIR is the theoretical spatial resolution. In microscopy it is physically impossible to have a resolution smaller than the wavelength of light used to probe a sample. This is called the diffraction limit. As the Raman process uses light of smaller wavelength (<1 micron) compared to the typical IR light (>1 micron) it allows in principle a higher resolution.

Complementarity

Far from being competitors, the two techniques which we have been writing about complement each other very nicely due to opposite selection rules in a lot of compounds. Indeed, we have seen that Raman-active modes are the ones that change in polarizability with the vibration.[2] This usually involves some kind of elongation or stretching of bonds making it easier for electron density to flow back and forth with the incident *E* field of light.

On the other hand, IR-active modes are the ones which create a dipole moment.[1] In many cases this can't be achieved by a symmetrical stretching motion which would not change the overall charge centres. Therefore Raman-active modes tend to be IR-inactive. Likewise, asymmetric stretching motions do not tend to change the molecular length and are thus usually Raman-inactive but IR-active because they create a dipole moment. In the case of centrosymmetric molecules, this is known as the **centrosymmetric rule.**[4] And this is directly applicable to CO_2 as we have seen. (Figure 10)

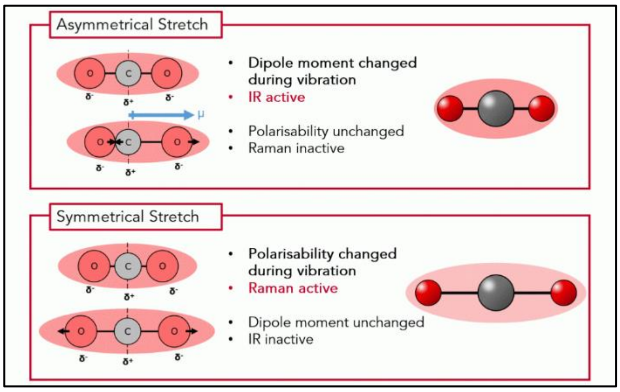

Figure 10: IR vs Raman selection rules for both kinds of linear stretches[1]

However, this complementarity doesn't work in all cases. Many modes can be both Raman and IR-active in systems like polar molecules or crystalline solids. In fact, the more symmetric a molecule is the more likely the two techniques will be complementary and vice-versa.[6]

Raman versus FTIR in practice

We will see in this section what kinds of materials can be studied with both techniques and in what fields might those analyses be relevant. We will also address how some theoretical limitations can be overcome with modified setups such as ATR-FTIR and enhanced-Raman techniques.

Raman or FTIR? A broad practical comparison

As both techniques are part of the vibrational spectroscopy family, we can broadly say that they give similar information about materials overall. These tools are incredibly versatile and allow scientists to study many kinds of crystalline or amorphous solids[2], polymers, ceramics, liquids, and gases, including their properties such as crystalline orientation, temperature, pressure... But some materials are nonetheless more suitable than others.

Raman vs FTIR in liquids

the Raman method was discovered by C.V.Raman in liquids. In aqueous solutions in particular Raman analysis is at a great advantage[2] compared to FTIR because water is a great absorber of IR light[1] at large wavenumber bands that are useful in chemistry to characterise other (mostly organic) molecules.
Aqueous solutions are omnipresent in a number of different fields: pharmaceutical[2], environmental science and most of all biology where it is also very effective for complex aqueous mixtures like biological fluids.[12]

It can also work as a microscopy technique because as we saw Raman has a better spatial resolution. Some Raman microscopes have a resolution close to that theoretical limit at <1 micron, which allows the precise study of cells and their complex molecules such as proteins or DNA. However, one must be careful not to concentrate too much laser power on a small area as that could possibly burn or damage the sample.

Though, FTIR can also be at an advantage in the broader world of organic chemistry because unlike it, Raman tools can't monitor fast reactions in real time. This is due – as we have seen – to the inherent weakness of Raman scattering inducing long acquisition times. FTIR is also better at identifying functional groups with high sensitivity which play a central role in organic chemistry.

To summarize it could be said that FTIR is better in this field if the solvent is dry.

Raman vs FTIR in the solid state

Raman spectroscopy is a powerful tool in crystallography. It can complement traditional methods like X-ray diffraction by providing detailed information about the symmetry, composition, and orientation of crystals on a micrometre scale. Unlike methods like X-ray scattering, Raman scattering provides information on inhomogeneities, local strain, lack of crystallinity, anharmonicities, or electronic structure specifics.[9]
Similarly in geology, it serves as a valuable technique for characterizing minerals and understanding geological formations. It aids in identifying mineral compositions and detecting polymorphs. It turns out that FTIR provides similar although less abundant information regarding crystals.

Due to Raman's nature as a scattering-based tool, it is well adapted in the analysis of thin films and coatings and all the applications that comes with them. We will see that ATR-FTIR is also well-suited for surfaces. Both tools enable researchers to map chemical composition but also measure layer thickness and assess structural integrity. Their non-destructive nature makes it ideal for examining delicate coatings without altering their properties. Additionally, Raman's high spatial resolution is yet again an advantage for these kinds of objects.

Conversely, Raman spectroscopy isn't very adapted to metallurgy due to a few limitations. One significant challenge is that metals often generate strong fluorescence signals that can overwhelm the Raman signals, making it harder to obtain accurate data. Furthermore, both tools require molecular bonds to operate.[2] Likewise, FTIR is not used in the analysis of metal bulk sample but can be to analyse contaminants and coatings without being subject to fluorescence interference.

In any case, other powerful techniques are already very well established and pertinent in the field of metallurgy like Scanning Electron Microscopy (SEM) or X-ray diffraction or scattering techniques like EDS.

Raman vs FTIR in the gas phase

The low sensitivity of Raman becomes worse in gases[7] where molecules are not as concentrated. In contrast, FTIR is used in gases to detect trace chemicals down to parts per billion.

Despite this, Raman analysis still holds a good advantage other FTIR for its ability to see vibrational and rotational transitions in simple gases like N_2, O_2, Cl_2, and other homonuclear diatomic molecules.[7] This is important for monitoring oxygen or nitrogen levels in various industries, pharmaceutical and medical analysis (anaesthetic and exhaled gases, monitoring gas impurities in the packing of drugs...),[7] and remote gas analysis in aerospace and combustion engines.

Because Raman analysis only rests on scattering and not absorption like FTIR, it has a unique capability to analyse samples remotely from a single side, enabling remote gas composition and temperature analysis in industrial chimneys and turbines (for industrial or law enforcement reasons).[7] Again, this application can be challenging due to signal weakness buried in the environmental noise meaning that FTIR is still very much in use in those same domains.

In geology, Raman gas analysis can be used to determine the composition and pressure of gas bubbles trapped in minerals[7] while FTIR is used to analyse gases coming out of volcanoes.

The major downside of both tools in gas phase is the inability to detect monoatomic species such as noble gases[11] because they don't vibrate.

Beyond theoretical limitations with inventive setups

Enhanced Raman techniques: SERS and TERS

In some cases, the weak Raman signals can be dramatically enhanced as well as its spatial resolution. Two groundbreaking advancements, Surface-Enhanced Raman Spectroscopy (SERS) and Tip-Enhanced Raman Spectroscopy (TERS), have propelled the capabilities of traditional Raman methods. SERS, discovered by accident in 1974[2] employs specially crafted surfaces, often made of metals like gold or silver, to intensify Raman signals, allowing detection at incredibly low concentrations, even down to a single molecule.[12] On the other hand, TERS uses nanoscale sharp metal tips to concentrate light, achieving ultra-high spatial resolution and enabling the study of materials at the nanometre scale. Thus, both techniques break barriers in sensitivity and resolution.[2]

In forensics, the sensitivity of SERS is used to analyse inks, blood, drugs residues and more.[12] The same ideas apply to archaeology which is in a way a kind of forensic science.

ATR-FTIR

In the very beginning of IR spectroscopy, the technique was a lot more limited in terms of what could be analysed. Because it is an absorption technique, light had to go through the sample, meaning that solid samples had to be made very thin: a very tedious operation that can't always be done. But then the ATR method was invented. It has now become omnipresent in FTIR setups.

ATR means Attenuated Total Reflectance. The principle is that instead of going through the whole sample, the light beam is reflected on the surface of the sample at a certain angle, while remaining technically an absorption technique. This is because upon reflecting on the sample, the light waves create a penetrating standing evanescent wave inside the first few microns of the sample surface. This evanescent wave interacts with the molecular vibrations in the same way as before.[23]

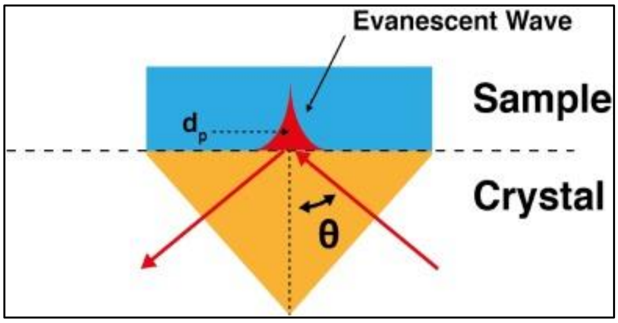

Figure 11: Path of a light beam in ATR

However, doing that yields a much weaker signal because the absorption is done over a much shorter distance. This issue is solved in ATR using a crystal below the sample that will internally reflect the beam multiple times onto the sample surface.[22]

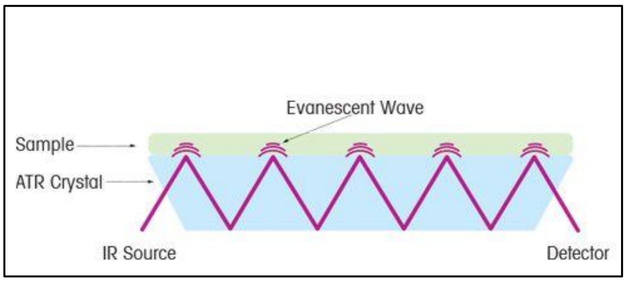

Figure 12: Multiple reflections setup in ATR

Thanks to this innovation, FTIR can now analyse a much wider class of materials such as liquids and gases like before, but also opaque and solid materials. ATR is the most widely used kind of FTIR spectroscopy, which is itself the most widely used technique of IR spectroscopy.

The biggest downside of ATR is the need to press the sample against a crystal before analysis, which requires either smooth flat materials or the application of pressure on the sample.

Results and discussion

In this comparative study between FTIR and Raman spectroscopies, the distinct theoretical underpinnings and practical applications of both techniques have been explained and compared.

Results

Theoretical considerations shed light on specific constraints and merits of these tools like their complementary yet sometimes overlapping selection rules, Raman's superior spatial resolution, and (ATR-)FTIR’s superior rapidity and sensitivity.

In practice, both techniques exhibit versatility in studying a wide array of materials, yet each excels in specific domains.
Raman spectroscopy is best for dealing with water-containing sample, crystallography, inorganic chemistry, and thin film investigations. We find that neither tool is generally pertinent in metallurgy compared to other more suited techniques. Although there is great overlap in what both vibrational techniques can study, we find that FTIR is still best put to use in dry organic chemistry and related fields due to good identification of functional groups but poor sensibility trough water.[1]
In the gas phase we find that that Raman’s capacity for remote chemical/thermal analysis of simple diatomic gases counterbalances FTIR’s great relative sensitivity to other gases. We explain that this can be put to good use notably to analyse combustion products.

Lastly, we briefly touched on the merits of enhanced Raman techniques like SERS and TERS, and how they have dramatically amplified Raman signals and spatial resolution, broadening applications in forensics and archaeology as well as why ATR-FTIR has been a cornerstone in FTIR spectroscopy for decades by allowing the simple analysis of opaque solids.

Discussion and ameliorations

Throughout this paper we have seen that such a practical comparison can indeed be somewhat traced back all the way to the very basics of Raman scattering versus IR absorption. However, it is important to acknowledge the many limitations and areas where the analysis could have been more comprehensive.

Indeed, the concise nature of this report restricted the depth of explanation, resulting in an overview that omitted several details. While the theoretical foundation of Raman and FTIR spectroscopy was outlined, the complexity of rotational and rovibrational transitions, as well as fluorescence interference and mitigation strategies in Raman were completely skipped.

In terms of the actual instruments, this study did not delve into the varied instrumental setups that one can find on the market nor build on that to explain in more depth enhanced Raman techniques, of which resonance Raman was not mentioned. Furthermore, the explanation of the Fourier Transform, a critical component of FTIR, was only provided at a surface level. There is also an absence of discussion regarding the precise capabilities and limitations of Attenuated Total Reflectance (ATR) compared to standard FTIR. Lastly, we overlooked ongoing advancements and emerging techniques in spectroscopy.

These many omissions greatly limited the exhaustive understanding of these techniques and their respective advantages in practical applications.

Conclusion

While FTIR is overall a more popular technique, both Raman and FTIR can yield significant advantages in some respects coming from the basic theory and translating into their popularity in various applications and fields.

However, it becomes evident through this report that the complexity and multitude of advancements in these techniques transcend the basic theoretical underpinnings offered here. This suggests the need for more comprehensive resources in selecting appropriate spectroscopic tools for specific applications.

In conclusion, while this report still provides a foundational comparative understanding of theory and applications, it underscores the necessity for more detailed and specialized resources to guide researchers effectively in the diverse and evolving landscape of spectroscopic techniques.

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